

Chapter

The Excretory System

1. The Loops of Henle lie in COMP 2021

- (a) Renal Cortex
- (b) Renal Medulla
- (c) Renal Pelvis
- (d) Renal Artery

2. The structure that stores urine temporarily is : COMP 2021

- (a) Ureter
- (b) Urethra
- (c) Urinary bladder
- (d) Kidneys

3. The main nitrogenous waste formed in the human body : SQP 2021

- (a) Uric acid
- (b) Urea
- (c) Ammonia
- (d) Creatinine

4. The principal nitrogenous waste in man is COMP 2022

- (a) urea
- (b) uric acid
- (c) ammonia
- (d) nitrogen

5. Ultrafiltrate generated by the glomerulus is having all the constituents of the blood plasma except MAIN 2023

- (a) protein
- (b) WBC
- (c) RBC
- (d) All of these

6. Excretion primarily involves MAIN 2021

- (a) removal of all byproducts during anabolism
- (b) removal of byproducts during anabolism
- (c) throwing out excess water
- (d) removal of nitrogenous wastes

7. Glomerular filtrate does not contain MAIN 2023

- (a) glucose
- (b) amino acids
- (c) hormones
- (d) proteins

8. More ADH means COMP 2021

- (a) less permeable DCT
- (b) dilute urine
- (c) concentrated urine
- (d) fall in blood osmotic pressure

9. The main nitrogenous waste in man. SQP 2023

10. Which substances are reabsorbed in distal convoluted tubules ?

MAIN 2021

11. State whether the following statements are True or False. If False rewrite the correct form of the statement by changing the first or last word only :
Urethra carries urine from kidney to the urinary bladder. COMP 2023
12. What is the purpose of tubular secretion? MAIN 2022
13. How is blood brought to Bowman's capsule ? SQP 2022
14. "Urine is formed from alkaline blood but it is acidic in nature". Explain the significance of the above statement. COMP 2023
15. The inner part of kidney is called____. MAIN 2023
16. State the difference between Micturition and Parturition. SQP 2025
17. Differentiate between Afferent arteriole and Efferent arteriole (diameter).
MAIN 2024
18. Renal artery carries blood to the kidney. MAIN 2023
19. Renal vein contains less urea. SQP 2021
20. There is frequent urination in winter than in summer. SQP 2023
21. Draw a neat, labelled diagram of a Malpighian Capsule SQP 2025
22. Draw a neat diagram of a longitudinal section of a human kidney and label Renal Cortex and Renal Medulla on the diagram. MAIN 2021
23. The excretory products formed by liver. MAIN 2022
24. Expand the abbreviation - ADH. Give one example of a diuretic substance.
MAIN 2022
25. Afferent Arteriole and Efferent Arteriole MAIN 2022
26. Mention the three major steps involved in the production of urine.
SQP 2021

Solution

1. (b) Renal Medulla

2. (c) Urinary bladder

The urinary bladder is a hollow, muscular organ that collects urine from the kidneys via the ureters and stores it until it is expelled through the urethra.

3. (b) Urea

4. (a) Urea

Urea is the main nitrogenous waste in humans, formed in the liver through the urea cycle.

5. (d) All of these

6. (d) Removal of nitrogenous wastes

Excretion is the biological process of removing metabolic waste products, especially nitrogenous wastes like urea, uric acid, and ammonia, from the body.

7. (d) Proteins

Proteins are too large to pass through the glomerular filtration membrane, so they are absent in the filtrate. Glucose, amino acids, hormones, and small molecules are present.

8. (c) Concentrated urine

Increased ADH (antidiuretic hormone) makes the distal convoluted tubule (DCT) and collecting ducts more permeable to water, leading to greater water reabsorption and the formation of concentrated urine.

9. urea

10. Water and sodium.

11. *Ureter* carries urine from the kidney to the urinary bladder.

12. Tubular secretion helps remove harmful substances that are not filtered by the glomerulus by discharging them into the urine.

13. The renal artery carries oxygenated blood containing nitrogenous wastes to the Bowman's capsule.

14. Blood is slightly alkaline with a pH of 7.3–7.45, but urine is acidic because the kidneys remove acidic waste products from the blood. These acidic substances are filtered and excreted into the urine, helping maintain the body's acid-base balance.

15. Medulla

16. **Micturition:** The process of expelling urine from the urinary bladder.

- **Parturition:** The process of expelling the fully developed foetus from the uterus at the end of gestation.

17. **Afferent arteriole:** Has a larger diameter and carries blood from the renal artery to the glomerulus.

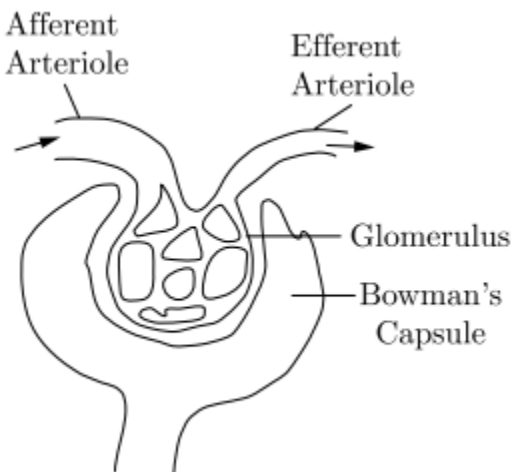
Efferent arteriole: Has a smaller diameter and carries blood away from the glomerulus to the renal vein.

18. True

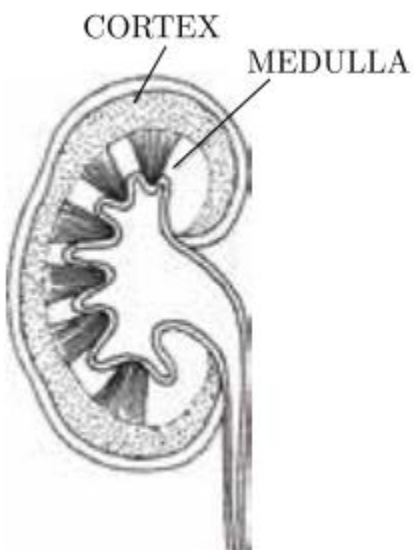
19. True.

20. In winter, sweating is reduced, so excess water is excreted through urine. In summer, more water is lost through sweating, leading to less urine formation.

21. Diagram of Malpighian capsule



22.



23. Urea and bile (which contains bile pigments).

24. ADH: Antidiuretic Hormone

Example of a diuretic substance: Alcohol, caffeine, or coconut water

25.

Afferent Arteriole	Efferent Arteriole
It carries blood to the glomerulus .	It carries blood away from the glomerulus .
It carries blood containing waste materials .	It carries blood having less waste material .

26.

1. **Ultrafiltration** – Filtration of blood in the glomerulus.
2. **Selective Reabsorption** – Reabsorption of useful substances into the blood.
3. **Tubular Secretion** – Secretion of wastes into the tubules to form urine.